

International Conference on Machine Learning and Data Engineering

Hybrid Deep Transfer Learning for Enhanced Brain Tumor Detection through the Integration of MobileNetV2 and InceptionV3

Roseline Oluwaseun Ogundokun^{1, 2, *}, Charles Awoniyi³, Nabeela T. Adebola⁴,
Akinrotimi Akinyemi⁵, Samson Adebisi Akinpelu⁶, Matthew O. Adigun⁷

¹Department of Centre of Real Time Computer Sciences, Kaunas University of Technology, Kaunas, 44249, Lithuania.

²Department of Computer Science, Landmark University Omu-Aran, Nigeria.

³Bupa Insurance, United Kingdom

⁴Data Science Department, University of Salford, United Kingdom

⁵Department Of Computer Science, Caleb University, Imota, Lagos, Nigeria

⁶School of Mathematics, Statistics and Computer Science, University of Kwazulu-Natal, Durban South Africa.

⁷Department of Computer Science, University of Zululand, KwaDlangezwa, South Africa.,

Abstract

Brain Tumors (BT) are one of the most hazardous diseases in the world, primarily because of the speed at which they spread and high mortality rates. Hence, diagnosing BT at an early stage with higher accuracy is necessary. Recent breakthroughs in deep transfer learning have achieved great performances in identifying common types of BT, such as glioma, meningioma, and pituitary tumors. In this paper, we will propose a novel hybrid deep transfer learning architecture by combining the strengths of two state-of-the-art pre-trained deep convolutional neural networks: MobileNetV2 and InceptionV3. We implement a parallel feature extraction process, reduce dimensionality in the extracted features using dropout techniques, and pass the refined data through fully connected layers for accurate classification. In this context, our two-step methodology of sub-image classification and final image categorisation with the help of majority voting and maximum probability principles shows much-improved results. Validated on a comprehensive BT dataset, our approach achieves 98.68% accuracy, 97% precision, 97.25% recall, and 97% F-measure with 0.9975 AUC and average precision of 99.50%. The outcomes surpass the performance of existing models on precision, sensitivity, and generalization, thus revealing the value that this technique may hold in whole-slide MRI image classification and clinical practice.

© 2025 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: Brain tumor; Transfer learning; Fine-tuning; Deep learning; Pre-trained model; Deep neural network

* Corresponding author. Tel.: +2347036261504

E-mail address: rosogu@ktu.lt; ogundokun.roseline@lmu.edu.ng

1. Introduction

The human brain is one of the most complicated organs, consisting of billions of cells. BT progresses as a result of uncontrolled growth of cells and can potentially form a mass of aberrant cells within or near the brain [1]. This aberrant cell mass has the potential to destroy normal cells of the brain and impede normal functions of the brain [2,3]. BT grades could also be divided into Grades I and II, which are benign or of low grade, and Grades III and IV, which are malignant. Brain benign tumors do not tend to transmit to further parts of the body; they are not cancerous, and their growth in the brain itself is prolonged. As such, malignant brain tumors grow fast and are poorly defined. They can either originate within the brain, known as primary cancer, or elsewhere in the body and then can be conveyed to the brain, known as secondary cancer. Brain Magnetic Resonance Imaging has been the primary tool most researchers adopt in detecting and characterizing brain cancers and modelling tumor growth during treatment. MRI generates high-resolution images that have formed the basis for automated medical image processing [4, 7, 9] and provides critical information regarding brain structure and tissue anomalies. The subsequent digitization of these medical images has led to the advancement of several automatic techniques for BT recognition and classification on top of MRI images [10].

However, current deep learning (DL) models have started a fascinating machine learning (ML) trend. Deep designs may express complicated connections effectively without using as many nodes as shallow structures like SVM and KNN. Because of this, several health informatics disciplines, including bioinformatics, medical informatics, and medical image analysis, are quickly advancing [8, 10, 11]. Researchers employ the meningioma, pituitary, glioma, and no tumor classes of the standard Kaggle Magnetic Resonance Imaging (MRI) BT classification dataset. Based on this dataset, we conduct in-depth investigations to examine the effectiveness of four DL techniques for classifying BT MRI scans using the DTL approach. The study employed MobileNet, InceptionV3, InceptionResNetV2, DenseNet169, ResNet101, and Xception for the automated identification and classification of BT utilizing a fine-tuned classification technique. Additionally, several methods are used to identify and categorize brain tumors that prohibit them from being automated. These methods use humanly described cancerous areas.

This research aims to create and verify a novel methodology for automatically identifying and categorizing brain tumors employing DTL strategies. Specifically, this involves incorporating parallel DCNNs and employing advanced classification methodologies to improve the identification process's accuracy and sensitivity, surpassing current approaches' capabilities.

The contributions of this study are described as follows:

- i. The study focuses on the critical need in clinical practice, where there is a vital requirement for automated brain tumor detection using DTL for glioma, meningioma, and pituitary tumors in their early stages.
- ii. The proposal of a new technique—TumourBlendNet—merges two parallel deep convolutional neural networks, including MobileNet and InceptionV3, for effectively extracting unique features present in the brain tumor dataset images.
- iii. Dropout techniques applied to the feature maps reduce dimensionality, improving efficiency for subsequent classification tasks in the TumourBlendNet model.
- iv. The study employs a two-step classification process involving sub-image and overarching image classifications, combining majority voting and maximum probability principles. It outperforms existing techniques in accuracy and sensitivity on the Brain Tumor dataset.

The paper then reviewed the related work on brain tumor detection, as presented in section 2. This was followed by a detailed explanation of the TumourBlendNet model, presented in section 3, the experimental setup, and a results comparison in section 4. Finally, section 5 concludes with a summary of the contributions and future research directions.

2. Related Works

DL has been employed in the medical industry [12, 13] for different disease classifications, such as breast cancer [14-16], skin cancer [17-19] and so on. This article discusses the technology used to identify brain tumors, which interests many researchers.

Salchin [20] offered a technique for brain cancer's early detection. MRI was used to find and classify cancers into different tumor kinds. For image categorization tasks, deep learning delivers comparably effective results. Consequently, our study combined the TensorFlow framework with the convolutional neural network methodology. We demonstrate that a similar job is outperformed by the speedier CNN approach with 91.66 per cent accuracy. In another contribution, Ranjbarzadeh et al. [21] proposed a robust and flexible technique for BT segmentation. This method speeds up the computation while overcoming the excessive complexity of cascade DL models. This CNN technique generates local and global characteristics in two different directions. Additionally, it improved BT segmentation accuracy, which is similar to the current model. The recommended method raises the WT mean scores by 0.9203, 0.9113, and 0.8726 points.

Kokila et al. [22] recommended a system for using MRI to identify brain cancer. This involves locating the tumor and grading, categorizing, and discovering it. Instead of making a unique model for each categorization task, this approach used a single brain MRI tissue model for multiple classification tasks. Multitask classification founded on CNN can detect and classify tumors. A CNN-based model was also used for brain tumor location, with 92% accuracy in brain tumor segmentation.

Gumae et al. [23] combined a standard extreme learning machine with brain tumor segmentation (RELM). This technique first processed the image to make it easier for the method to evaluate. The system used a min-max strategy for preprocessing. The contrast of the raw image was significantly improved using the min-max preprocessing procedure.

Kaplan et al. [24] initiated BT recognition and segmentation by integrating two novel techniques: the first technique, called neighbour-distance-based LBP, considers the relationship of the neighbouring pixels, and the second technique, called LBP, considers the angle between the adjacent pixels. These methods were applied to preprocess the most frequent categories of BT. Character representation used histograms of preprocessed images, while the revised approach performed better than traditional ways of feature extraction employed by the model.

3. Methodology

This is a new, innovative concept for MRI image classification in this work, proposing the most advanced deep transfer learning techniques. Our model effectively combines two conventional pre-trained CNNs called MobileNetV2 and InceptionV3, custom-tuned for feature extraction. Unlike other classical models, we adopt only backbone feature extraction layers from both CNNs and discard the last fully connected layer. This enables the network to produce dense, down-sampled feature maps and rich representations of the input MRI images. Further, such produced feature maps undergo refinement using a Global Average Pooling-GAP layer that reduces their dimensionality while optimally preparing the data for the subsequent classification layers. The output from both models is combined, dropout is applied to avoid overfitting, and the refined data is fed to a final dense layer for robust classification.

- i. Our approach becomes prominent for the following novelty in steps:
- ii. Preprocessing: Normalization and enhancement of MRI images
- iii. Image augmentation to ensure the diversity of the dataset for generalization of the model
- iv. Extract features parallel to two CNN backbones to ensure comprehensive feature learning.
- v. Sub-image categorization followed by final image categorisation based on majority voting and maximum likelihood principles for enhanced accuracy and stability.
- vi. The integrated approach presented here is a very efficient system for MRI-based classification of a brain tumor, quite different from any other approaches presented so far.

The schematic diagram in Figure 1 illustrates the suggested approach in block format. Subsequent sections will provide a comprehensive breakdown of each phase.

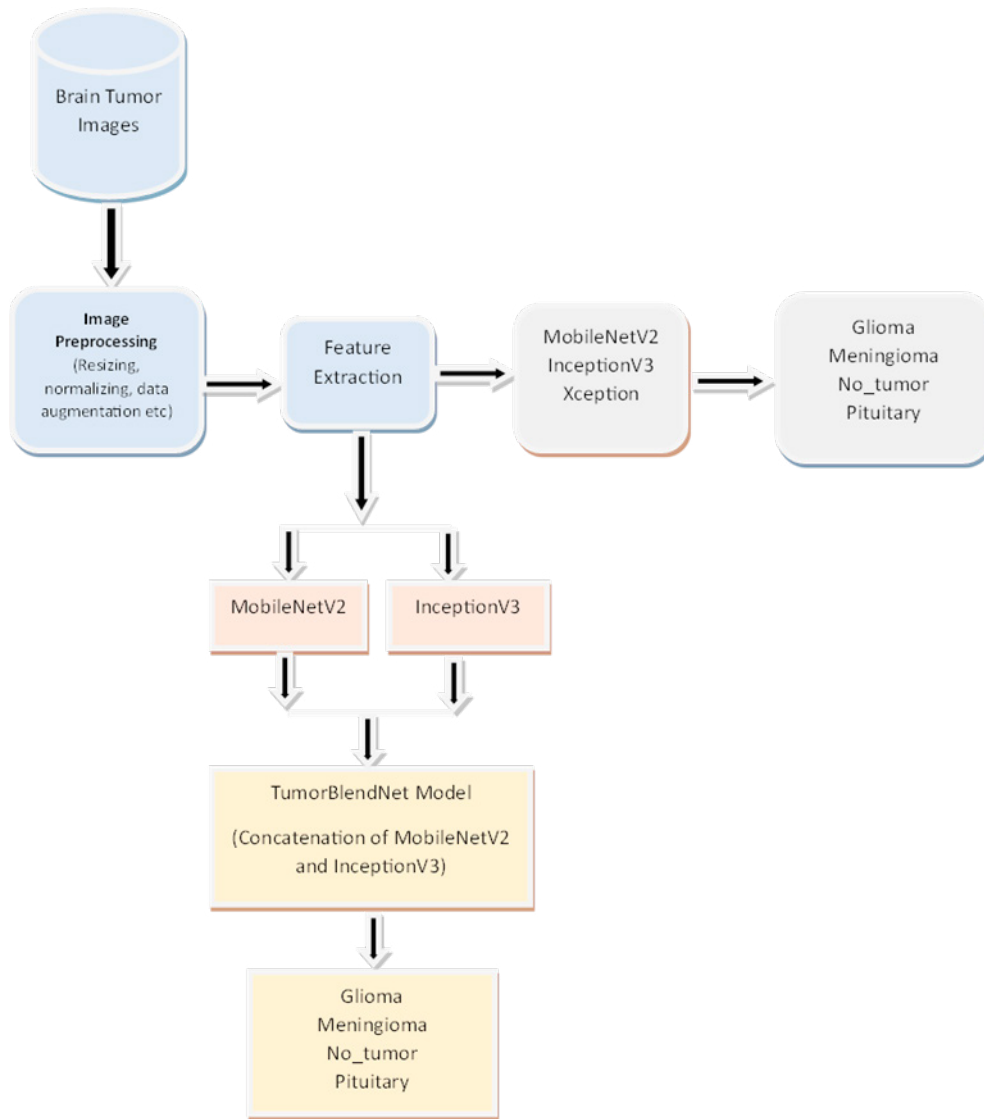


Fig. 1. Flow Diagram of Suggested Approach

3.1. Preprocessing of Dataset

Data sets were gathered from a collection of MRI brain images (JPGs) to identify brain tumours in a manner that matched the model applied to get the intended outcomes. The first task in data preprocessing was to scale the MRI brain images. Due to different image sizes, the image has been scaled down from 224 to 224 pixels. All photos are standardized according to the ImageNet standard. The image sample was then divided into four groups: glioma, meningioma, no_tumor, and pituitary, and uploaded to the system's local disk. We confirm it is loaded correctly and precisely utilizing Python code in the Jupiter Notebook. This dataset contains 7023 MRI scans, including 1621 glioma-positive patient scans, 1645 meningioma-positive patient scans, 1757 pituitary-positive patient scans, and 2000 brain tumor-negative patient scans. The dataset combines three data, including figshare, SARTAJ, and Br35H datasets. The no_tumor class was retrieved from the Br35H dataset. Table 2 gives an example of each category. Details of this split are presented in Table 1.

Table 1. Dataset Distribution.

Sets	Total Image
Training	5780
Validation	1020
Testing	1200
Total	8000

3.2. Data Augmentation

The data set has an unbalanced distribution of images between the four classes. An imbalance in the data set can cause objections to the majority layer. There is a way to expand the data in the dataset to avoid the problem of class imbalance. Each image is randomly rotated, cropped, resized, and altered to prevent whole image duplication. Data augmentation increased the volume of data, increasing the recommended model's competence to withstand feature changes. Various investigators have extensively explored the significance of data augmentation in DL since it often facilitates the handling of substantial amounts of data and the creation of generic algorithms without altering the image class [25-27]. Table 2 displays the distribution of classes within the picture collection before and after using data augmentation techniques.

Table 2. Distribution of the Data Augmentation.

Class	Before Augmentation	After Augmentation
Glioma	1621	2000
Meningioma	1645	2000
No_tumor	2000	2000
Pituitary	1757	2000
Total	7023	8000

3.3. TumourBlendNet Model

The TumourBlendNet model was used to modify the overall research approach of this study. The first stage in this approach is the data-gathering phase (Figure 1), when data is collected following the research study. The second step is the data preprocessing action, which combines, cleans, and transforms the collected information to make the data set suitable for the classification process. The data sets from the preprocessing stage (stage 2) were then transferred to stage 3 for the classification procedure. In Step 4, a model based on the fine-tuning process was designed with improvements, trained, and tested against data for a classification procedure. At the final stage of their study, the authors compare three DTL models: MobileNet, InceptionV3, and Xception.

4. Results and Discussion

The following section discusses in detail the dataset used for studying the BT classification implementation procedure—that is, the configuration of DTL—along with an in-depth analysis of results from various experiments

intended to evaluate the efficacy of our technique. Details about building the DTL techniques and the research's software platform may be found under the implementation configuration.

4.1. Implementation Parameter

This work uses architectures like pre-trained Transfer Learning, MobileNet, InceptionV3, and Xception. The categorization of BT using MRI scans is the focus of this work. Additionally, we used a trial-and-error method. Implementation was conducted to find the most suitable settings for every parameter by giving the parameters variables. We developed pretrained DL techniques using stochastic gradient descent (SGD) through DTL. We used a 32-image minibatch size and a 0.01 learning rate.

Furthermore, each DL technique has been trained for 20 epochs to mitigate overfitting and undertake the DTL tests for determining and classifying different brain tumors. Table 3 explains the best settings used during the identification process.

Table 3. The DTL Technique Parameters

Parameters	Values
Verbose	1
Optimizer	SGD
EarlyStopping	Patience 10
Epochs	20
Batch size	32
Learning rate	0.01
Regularisation	0.01

4.2. Implementation Results

The TumourBlendNet model recommended 5-layer method which provides excellent results in tumour detection. Five-layer CNN models with Convolution, Max Pooling, Flattening, and two dense layers have been presented. Due to CNN's transformation invariance, data augmentation occurred before the model was fitted. The performance was assessed by splitting the data set into three equal parts (80% TRA data set, 10% VAL data set, and 10% TES data set). As revealed in Table 4, the TumourBlendNet model realized the highest training, validation, and testing accuracies of 99.98%, 96.57%, and 97.00%, respectively, followed by MobileNetV2 with 99.88% training accuracy, 96.25% and validation accuracy. The inceptionV3 model also had a testing accuracy of 96% but had the lowest training and validation accuracies of 99.80% and 95.00%, respectively. At the same time, the TumourBlendNet model got the lowest train, validation, and test loss of 0.0011, 0.1229, and 0.1189, respectively, followed by the MobileNetV2 model having a train, validation and test accuracy (Acc.) of 0.0013, 0.1452 and 0.1358 respectively. It was demonstrated that MobileNetV2 and InceptionV3 models had a testing accuracy of 96.00%, while Xception had the lowest test accuracy of 95.00%.

This is great because it uses a 5-layer CNN. Although the studies were examined using varying numbers of layers, the difference in results was not significant using this 5-layer CNN model. Some aspects gained when increasing the number of layers were computational time, method complexity, batch size, and the number of steps. Furthermore, 0.2 was utilized as the dropout value. However, its accuracy did not match the model. With a drop of 0.4, this model thus offers the highest level of accuracy.

Table 4. Training and Validation Accuracies and Losses

Model	Train Acc.	Validation Acc.	Test Acc.	TRA loss	VAL loss	TES loss
TumorBlendNet	99.98	96.57	97.00	0.0011	0.1229	0.1189
MobileNetV2	99.88	96.25	96.00	0.0013	0.1452	0.1358
InceptionV3	99.80	95.00	96.00	0.0017	0.1980	0.2382
Xception	99.85	96.08	95.00	0.0014	0.1549	0.2262

One of the core strengths of TL procedures and fine-tuning relates to the decrease of overfitting problems. Most of the time, DL methods usually suffer from overfitting in limited training, validation, and test data. The BT classification was executed based on the training and testing of the TL approaches using the same TL parameters described in Table 3. Figure 2 shows the Acc. and loss graphs for the TumorBlendNet approach, and it was found that TumorBlendNet worked best without detecting any overfitting.

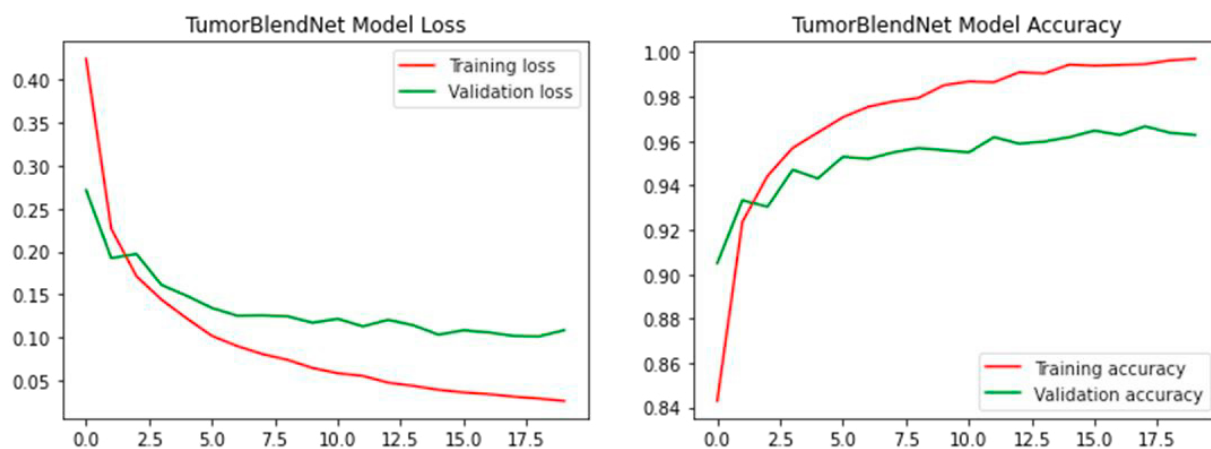


Fig. 2. TumourBlendNet Model Acc. and loss graphs for all four DTL Techniques

The confusion matrix further supported the TumourBlendNet Model, which performed exceptionally well, and the results. Table 4 details the performance of four DTL approaches in classifying BT MRI images. Results indicated that TumorBlendNet DTL had an average Acc. of 98.68%, followed by Xception with an average Acc. of 97.50%. At position three came MobileNetV2 with an average Acc. of 98.13%. InceptionV3 took the lead at 98.00%. Other deep neural network models held a more moderate ranking in terms of classification performance. In this research, each model created in the DTL was applied to the testing dataset, and the confusion matrix was developed to determine the effectiveness of the entire system. A crystal clear comparison of the TumourBlendNet with those presented in the earlier published works is best given in tabular format, as seen in Table 5. Most of the measures used to evaluate the TumourBlendNet model implemented gave maximum values compared to the already used models like MobileNetV2, InceptionV3, and Xception. The highest observed values were 98.68% for accuracy, 97.00% for precision, and 97.25% for recall. Emphasizing such improvements will signify that TumourBlendNet overcomes the weaknesses of previous models that suffered from reduced accuracy and were prone to false positives for brain tumor classification and is better suited for clinical applications. This structure would explain why the TumourBlendNet is more suitable for clinical applications.

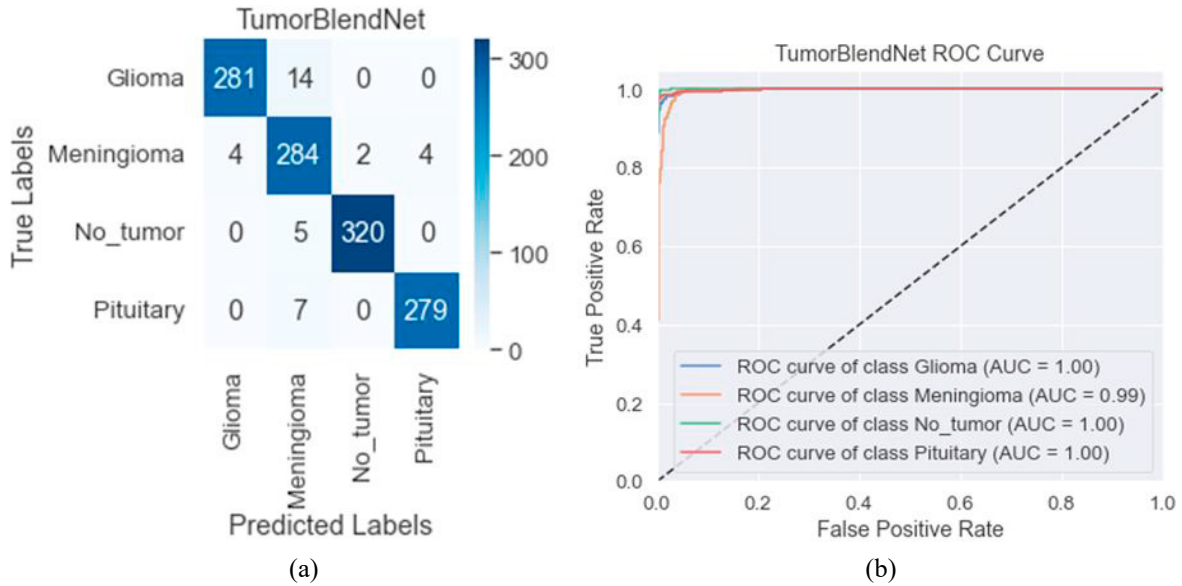


Fig. 3. (a) Confusion matrix of TumorBlendNet ; (b) ROC Curve for TumorBlendNet

Table 5 shows that the accuracy rises with model depth for deeper architectures like InceptionV3 since a deeper DL-based approach incorporates additional intricate and crucial deep characteristics and enhances the architecture's classification ability. Furthermore, as the architecture's depth increases, the computational cost also rises, which decreases the network's effectiveness. In addition, Table 5's findings show that the TumourBlendNet DTL technique is the most efficient way to identify and categorize brain tumors. Table 5 demonstrates that the TumourBlendNet model outperformed the remaining three DTL models, yielding an Acc. of 98.68%, a Pre. of 97.00%, Rec. of 97.25%, F-measure of 97.00%, AUC of 0.9975, and with a count of 1164 TPs. At a close range was the MobileNetV2 model, which had an Acc. of 98.13%, Pre. of 96.25%, Rec. of 96.25%, F-measure of 97.00%, with an AUC of 0.9975, AP of 0.9875, and a total of 1155 TPs. Good performance was observed in the case of Xception with an Acc. of 97.50%, Pre. of 94.75%, Rec. of 94.75%, F-measure of 94.75%, with an AUC of 0.9975, AP of 0.9850, and a total of 1140 TPs. It has been discovered in Table 5 that TumourBlendNet achieves the highest accuracy of 98.68%, F1-score, precision, and recall compared to MobileNetV2 InceptionV3 and Xception. Also, all the models develop similar AUC values out of which TumourBlendNet develops overall superior performance, proving to be very effective for brain tumor classification in clinical applications.

Table 5. Average Performance Metrics of the System

Model	Acc. (%)	Pre. (%)	Rec. (%)	F-measure (%)	AUC	AP	True Positive (TP)
TumorBlendNet	98.68	97.00	97.25	97.00	0.9975	0.9950	1164
MobileNetV2	98.13	96.25	96.25	96.50	0.9975	0.9875	1155
InceptionV3	98.00	96.25	96.00	96.00	0.9925	0.9850	1152
Xception	97.50	94.75	94.75	94.75	0.9975	0.9850	1140

The key performance metrics of various models deployed for brain tumour classification follow. The CNN model with an attention mechanism proposed by Ranjbarzadeh et al. resulted in minimum accuracy and F1-score of 92.03 and 91.58%, respectively, which was still entirely satisfactory given the challenging nature of the underlying task. Both

models deployed a hybrid CNN-LSTM approach, where Sharma et al.'s model outperformed the model of Rasool et al. marginally in terms of accuracy and F1-score: 97.20% vs. 96.79%, 96.85% vs. 96.16%, respectively. This showed the strength of the combination of CNNs with LSTMs. The results also indicated that TumorBlendNet model achieved the best performance for all metrics, including an accuracy of 98.68%, precision of 97.00%, recall of 97.25%, F1-score of 97.00%, and a significant AUC of 0.9975. This reflects that the brain tumor classification of the TumorBlendNet is more powerful and robust than the previously proposed models.

Table 6. Comparative Analysis with Existing Studies

Authors	Models	Accuracy	Precision	Recall	F1-score	AUC
Ranjbarzadeh et al. [28]	CNN with Attention Mechanism	92.03	91.13	92.03	91.58	0.9400
Rasool et al. [29]	CNN-LSTM	96.79	95.90	96.42	96.16	-----
Sharma et al. [30]	CNN-LSTM	97.20	96.80	96.90	96.85	0.9800
Proposed Model	TumorBlendNet	98.68	97.00	97.25	97.00	0.9975

5. Conclusion

In the presented work, a deep transfer learning-based BT identification model was investigated for detailed evaluation to automate BT classification using MRI data. DTL techniques were applied to classify the data into four categories: TumorBlendNet, MobileNetV2, InceptionV3, and Xception models. The results of our study were that the highest accuracy rate, 98.68%, belonged to TumorBlendNet. This would reveal that it had a high efficiency in correctly identifying and classifying BT MRI images. Though the study gave excellent results with a small dataset, further improvements could be developed by augmenting data to have more images in the training dataset. Even with the successes associated with this study, there is still room for additional work. The shortcomings observed in the independent performance of pre-trained deep learning architectures and the considerable time consumed while training transfer learning models suggest that refinements of these models with larger datasets should be the focus of future research. Additionally, further investigations of temporal latency for other competent DTL models could be studied to improve their tumor identification and classification. Advanced segmentation techniques can also be explored further to enhance the efficiency and effectiveness of our proposed approach.

References

1. Ajala, F. A., Akande, N. O., Adeyemo, I. A., and Ogundokun, R. O. (2017) "Smallest univalve segment assimilating nucleus approach to brain MRI image segmentation using fuzzy C-means and fuzzy K-means algorithms." *International Journal of Computers and Technology*, **16**(7), 7065-7076.
2. Kavitha, A. R., Chitra, L., and Kanaga, R. (2016) "Brain tumor segmentation using genetic algorithm with SVM classifier." *Int J Adv Res Electr Electron Instrum Eng*, **5**(3), 1468-1471.
3. Logeswari, T., and Kaman, M. (2010) "Improved implementation of brain tumor detection using segmentation based on a hierarchical self-organizing map." *International Journal of Computer Theory and Engineering*, **2**(4), 591.
4. Khambhata, K. G., and Panchal, S. R. (2016) "Multiclass classification of a brain tumor in MR images." *Int. J. Innov. Res. Comput. Commun. Eng*, **4**(5), 8982-8992.
5. Kaur, G., and Rani, J. (2016) "MRI brain tumor segmentation methods-a review."
6. Das, V., and Rajan, J. (2016) "Techniques for MRI brain tumor detection: a survey." *Int J Res Comput Appl Inf Tech*, **4**(3), 53e6.
7. Zacharaki, E. I., Wang, S., Chawla, S., Soo Yoo, D., Wolf, R., Melhem, E. R., and Davatzikos, C. (2009) "Classification of brain tumor type and grade using MRI texture and shape in a machine learning scheme." *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, **62**(6), 1609-1618.

8. R. O. Ogundokun, A. T. Abdullahi, A. R. Adenike, C. Awoniyi, H. B. Akande, F. Falajiki & V. Adedayo-Ajayi, "Hybrid Deep Learning for Breast Cancer Diagnosis: Evaluating CNN and ANN on BreakHis v1 400X," In 2024 International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG), April 2024, pp. 1-6. IEEE.
9. Singh, L., Chetty, G., and Sharma, D. (2012, November). "A novel machine learning approach for detecting brain abnormalities from MRI structural images." In *IAPR international conference on pattern recognition in bioinformatics* (pp. 94-105). Springer, Berlin, Heidelberg.
10. Pan, Y., Huang, W., Lin, Z., Zhu, W., Zhou, J., Wong, J., and Ding, Z. (2015, August) "Brain tumor grading based on neural networks and convolutional neural networks." In *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (pp. 699-702). IEEE.
11. Ravi, D., Wong, C., Deligianni, F., Berthelot, M., Andreu-Perez, J., Lo, B., and Yang, G. Z. (2016) "Deep learning for health informatics." *IEEE journal of biomedical and health informatics*, **21**(1), 4-21.
12. Adebisi, O.A., Ajagbe, S.A., Ojo, J.A., and Oladipupo, M.A. (2022). Computer Techniques for Medical Image Classification: A Review. In: Chakraborty, C., Khosravi, MR (eds) *Intelligent Healthcare*. pp 19–36, Springer, Singapore. https://doi.org/10.1007/978-981-16-8150-9_2
13. Kumar, K.S., Abirami, K., Sulochana, C.H., Kumar, T.A., Ajagbe, S.A., and Morris, C. (2024) "Multimodal Spatial-Temporal Prediction and Classification Using Deep Learning." In: A. J., Abimannan, S., El-Alfy, ES.M., Chang, YS. (eds) *Spatiotemporal Data Analytics and Modeling. Big Data Management*. Springer, Singapore. https://doi.org/10.1007/978-981-99-9651-3_5
14. Awotunde, J. B., Sisodia, D. S., Ayodele, P. A., Ogundokun, R. O., and Chouhan, V. S. (2022, October) "Diagnosis Expert System on Breast Cancer Using Fuzzy Logic and Clustering Technique." In *International Conference on Information Systems and Management Science* (pp. 589-601). Cham: Springer International Publishing.
15. Folorunso, S. O., Awotunde, J. B., Rangaiah, Y. P., & Ogundokun, R. O. (2024). "EfficientNets transfer learning strategies for histopathological breast cancer image analysis. *International Journal of Modeling, Simulation, and Scientific Computing*, **15**(02), 2441009.
16. Adelodun, A. B., Ogundokun, R. O., Yekini, A. O., Awotunde, J. B., and Timothy, C. C. (2023). Explainable artificial intelligence with scaling techniques to classify breast cancer images." In *Explainable Machine Learning for Multimedia Based Healthcare Applications* (pp. 99-137). Cham: Springer International Publishing.
17. Ogundokun, R. O., Li, A., Babatunde, R. S., Umezuruike, C., Sadiku, P. O., Abdulahi, A. T., and Babatunde, A. N. (2023) "Enhancing skin cancer detection and classification in dermoscopic images through concatenated MobileNetV2 and xception models." *Bioengineering*, **10**(8), 979.
18. Dildar, M., Akram, S., Irfan, M., Khan, H. U., Ramzan, M., Mahmood, A. R., ... and Mahnashi, M. H. (2021) "Skin cancer detection: a review using deep learning techniques." *International journal of environmental research and public health*, **18**(10), 5479.
19. Nahata, H., and Singh, S. P. (2020) "Deep learning solutions for skin cancer detection and diagnosis." *Machine learning with health care perspective: machine learning and healthcare*, 159-182.
20. Salçin, K. (2019) "Detection and classification of brain tumors from MRI images using faster R-CNN." *Tehnički glasnik*, **13**(4), 337-342.
21. Ranjbarzadeh, R., Bagherian Kasgari, A., Jafarzadeh Ghouschi, S., Anari, S., Naseri, M., and Bendeche, M. (2021) "Brain tumor segmentation based on deep learning and an attention mechanism using MRI multi-modalities brain images." *Scientific Reports*, **11**(1), 1-17.
22. Kokila, B., Devadharshini, M. S., Anitha, A., and Sankar, S. A. (2021, May) "Brain tumor detection and classification using deep learning techniques based on MRI images." In *Journal of Physics: Conference Series* (Vol. 1916, No. 1, p. 012226). IOP Publishing.
23. Gumaei, A., Hassan, M. M., Hassan, M. R., Alelaiwi, A., and Fortino, G. (2019) "A hybrid feature extraction method with regularized extreme learning machine for brain tumor classification." *IEEE Access*, **7**, 36266-36273.
24. Kaplan, K., Kaya, Y., Kuncan, M., and Ertunç, H. M. (2020) "Brain tumor classification using modified local binary patterns (LBP) feature extraction methods." *Medical hypotheses*, **139**, 109696.
25. Ogundokun, R. O., Maskeliūnas, R., and Damaševičius, R. (2022) "Human Posture Detection Using Image Augmentation and Hyperparameter-Optimized Transfer Learning Algorithms." *Applied Sciences*, **12**(19), 10156
26. Huh, M., Agrawal, P., & Efros, A. A. (2016) "What makes ImageNet good for transfer learning?" *arXiv preprint arXiv:1608.08614*.
27. Baghdadi, N. A., Malki, A., Balaha, H. M., AbdulAzeem, Y., Badawy, M., and Elhosseini, M. (2022) "Classification of breast cancer using a manta-ray foraging optimized transfer learning framework." *PeerJ Computer Science*, **8**, e1054.
28. Ranjbarzadeh, R., Bagherian Kasgari, A., Jafarzadeh Ghouschi, S., Anari, S., Naseri, M., and Bendeche, M. (2021) "Brain tumor segmentation based on deep learning and an attention mechanism using MRI multi-modalities brain images." *Scientific Reports*, **11**(1), 1-17
29. Rasool, M., Ismail, N. A., Boulila, W., Ammar, A., Samma, H., Yafooz, W. M., and Emara, A. H. M. (2022) "A hybrid deep learning model for brain tumour classification." *Entropy*, **24**(6), 799.
30. Sharma, P., Gupta, R., Sharma, S., Kotru, M., Singh, G., and Sikka, M. (2023) "Coagulopathy in the absence of overt DIC in postoperative neurosurgical patients is a strong predictor of poor outcome." *American Journal of Blood Research*, **13**(4), 110.